9	**(a)**		$m = \frac{3.58-3.93}{4-0}$ $\log_{10}V = -0.0875t + 3.93$	**M1** **M1** **A1** **[3]**	**1.1** **3.3** **1.1**	(±) 0.0875 oe Allow any value for – 0.0875 condone omission of base	Condone ln
9	**(b)**		$V = 10^{-0.0875t + 3.93}$ $A = 10^{3.93} = 8500$ $b = 10^{-0.0875} = 0.82$	**M1** **A1** **A1** **[3]**	**3.1b** **3.3** **1.1**	Or $\log V = \log A + \log b^t$ Accept $V = 8500\times0.82^t$	 8511.38 0.81752; sc B1 if both answers not to 2 sf. Allow 8500 taken from table (M1A1)
9	**(c)**		3151.29 (≈ 3150) so the model is a (very) good fit	**E1** **[1]**	**3.4**	Must see value from model OR find value of t (approx. 4.98) when model gives 3150	Eg 3107.93 from more accurate values.
9	**(d)**		$8500\times0.82^t = 500$ soi take logs of both sides (t = 14.27 so) after 15 years	**M1** **M1** **A1** **[3]**	**3.1b** **1.1** **3.5a**	Ft from (b) for M1M1 eg $\log8500 + \log0.82^t = \log500$; can be implied by 15 or 14.(…)	or $\log0.82^t = \log(\frac{500}{8500})$